

Overflow / wrap

Fixed-width math does not grow a bigger box when the answer is too big

Wrap versus signed overflow

- Unsigned wrap is modular arithmetic: results live modulo two to the width
- Add fourteen and three in four bits and you store one, not seventeen
- Signed overflow is different
- Carry and signed overflow are not the same question

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Overflow / wrap-around

Watch fixed-width add/sub **wrap mod 2^w** — unsigned modular wrap vs signed overflow hazards.

Starter example: Width 4, unsigned $14 + 3$. True sum 17 wraps to 1 (mod 16). Switch to signed and try $7 + 1$ for overflow.

Load starter example

Challenges 0 / 22 cleared

Quiz: mod: Wrapped result uses modulus? Answer: 2^w

Answer

Show hintCheckNextIdle

Quiz: modQuiz: flagsQuiz: satQuiz: counterDemo unsignedDemo signedAdd in rangeSub wrapSigned OK

Width 8Quiz: 14+3Quiz: 7+1 signedMax + 1Inc wrapMode signedQuiz: u maxQuiz: s maxBoth ops

Same bits different storyQuiz: HDLSigned sub ovFull hazard

Core ideas

Wrap

Bits always compute mod 2^w . Unsigned “overflow” is modular wrap.

Signed overflow

Result bits exist, but the signed meaning left the representable range.

Overflow wrap starter

learn_digital — overflow wrap

Workbook practice

- In the workbook track, pick width four
- On paper, add fourteen plus three unsigned and show why the stored result is one
- Treating carry as if it meant signed overflow

Pitfalls to watch

- Do not assume saturation, clamping to min or max, is the same as wrap
- Do not ignore width when you “check the answer in your head” with unlimited decimal
- And remember: the browser lab is literacy
- Waveforms still show the wrapped bits, not the mathematical ideal

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, work one unsigned wrap and one signed overflow example
- When you are ready, take the short quiz, then continue to ASCII and hex dump literacy